

WASP-43b

R-2636

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_140118 · created 2026-08-29T00:53:57+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456675.86824 +/- 0.00068 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.018
Transit depth (Rp/Rs)^2	2.07 +/- 0.52 [%]
Orbital Inclination (inc)	87.05 +/- 2.03
Airmass coefficient 1 (a1)	0.971 +/- 0.013
Airmass coefficient 2 (a2)	0.021 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.32 %
Best Comparison Star	#1 - [88, 394]
Optimal Method	PSF photometry
Transit Duration (day)	0.059 +/- 0.0026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.144 ± 0.018 vs 0.1594 (deviation 0.86σ)
Duration fit vs published	0.059 d vs 0.0512 d (deviation 2.98σ)
Mid-time O-C	1.5 min
Depth SNR	8.0
Residual scatter	1.32 %

Light curve

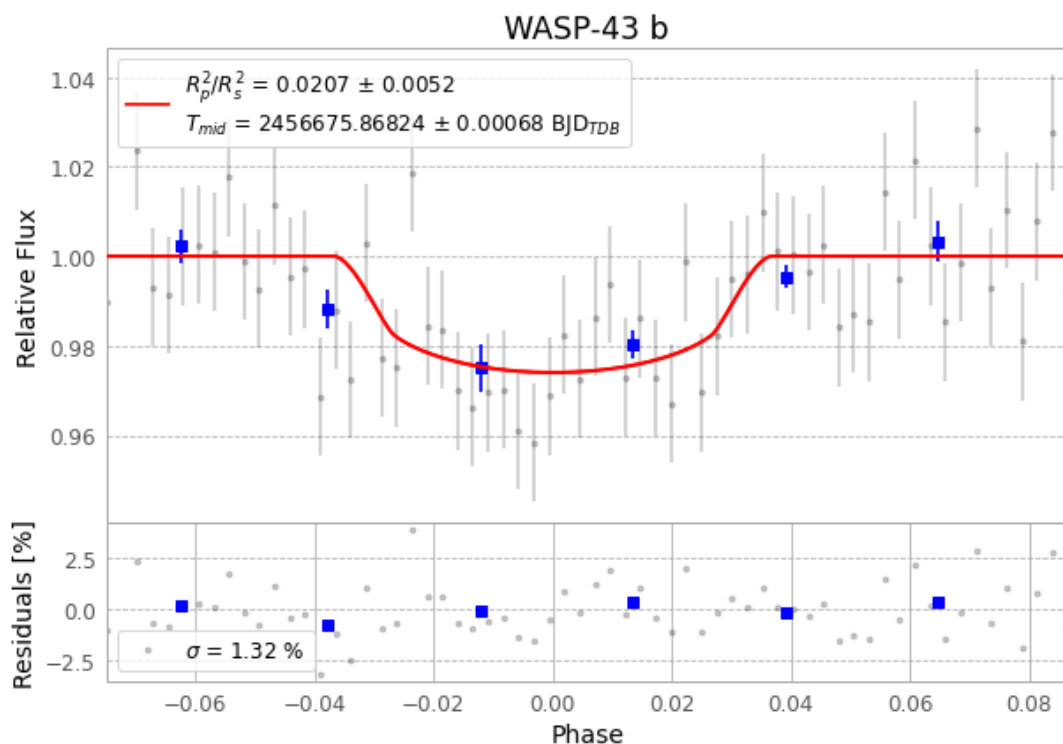


Figure 1 — FinalLightCurve_WASP-43 b_2014-01-18.png (SHA-256 1e0ca77f52cf7514...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [252, 218], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d6ee875cc065cb04...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43140118071512.FITS → WASP-43140118102412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140118.log (0ff84e104248...), FinalLightCurve_WASP-43 b_2014-01-18.png (1e0ca77f52cf...), FinalLightCurve_WASP-43 b_2014-01-18.csv (3a20368f79ac...)

Compute resources

Wall time 118.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 00:53Z.